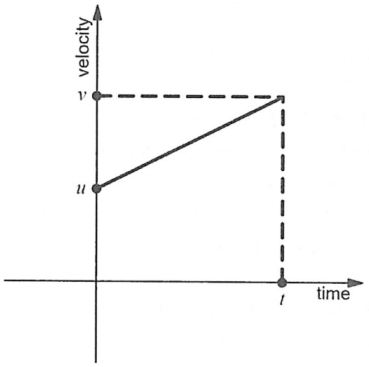
Constant Acceleration

Questions 1–32

Question 1. Using the following velocity-time graph



show that

a) $v = u + at$
b) $s = \frac{1}{2}(u + v)t$
c) $s = ut + \frac{1}{2}at^2$
d) $s = vt - \frac{1}{2}at^2$

Question 2. By substituting $v = u + at$ into $s = vt - \frac{1}{2}at^2$ show how $v^2 = u^2 + 2as$ may be derived.

Question 3. In each case, use the constant acceleration formulae with the given values:

a) $a = 2, u = 4, v = 6$, to find t
b) $a = -2, t = 3, s = 5$, to find u
c) $s = 12, t = 4, v = 5$, to find a
d) $u = 2, a = 1.5, v = 3$, to find s
e) $s = 20, t = 8, u = 5$, to find v

Question 4. If $s = 20, t = 6, v = 2.5$, using the constant acceleration formulae, find u and a

Question 5. If $s = 15, v = 4.5, u = 1.5$, using the constant acceleration formulae, find a and t

Question 6. A particle is moving in a straight line from A to B with constant acceleration 2 ms^{-2}. Its speed at A is 6 ms^{-1} and it takes 7 seconds to travel between A and B. Find

a) the speed of the particle at B
b) the distance from A to B

Question 7. Katie is walking along a straight path of length 100 m initially at 1.1 ms^{-1} and has constant acceleration 0.05 ms^{-2}. Find the time it takes for Katie to travel the path.

Question 8. Fasil is running along the road and for this part he can be modelled as running with constant acceleration of 0.1 ms^{-2}. He is running at 5 ms^{-1} when he passes a lamp post. A further 10 seconds later he passes a litter bin. Find

a) the distance between the lamp post and the litter bin.
b) the speed at which Fasil passes the litter bin.

Question 9. Find the acceleration and the displacement given each of the following scenarios, where the object starts from rest:

a) $v = 4$ ms^{-1}, $t = 3$ s
b) $v = 1.4$ cms^{-1}, $t = 2$ s
c) $v = 10$ ms^{-1} giving your answer in terms of t

Question 10. A cyclist is cycling on a straight flat road. Her initial velocity is u ms^{-1}, then she accelerates at a constant rate of a ms^{-2} for 12 seconds before reaching a speed of $3u$ ms^{-1}.
Express a in terms of u

Question 11. In perfect conditions, it takes 38 metres to stop a car driving at a speed of 80 kmh^{-1}.

a) Find the time it takes to stop assuming the deceleration is constant.

b) Comment on why this may not be an accurate answer for modelling the stopping of a car in a real-world scenario.

Question 12. Ceri cycles 2 km on a straight horizontal section of a race. She is cycling 28 kmh^{-1} initially and finishes the section at 32 kmh^{-1}.
Find her acceleration for this section, assuming she accelerates at a constant rate.

Question 13. A car travels along a straight horizontal road with constant acceleration, passing three points A, B and C. It takes 15 s to travel from B to C. It is moving at 12 ms^{-1} and 16 ms^{-1} at B and C respectively.

a) Find the acceleration of the car

b) Find the speed of the car at A given the distance between A and B is 150 metres.

Question 14. A car is travelling along a straight horizontal road with a constant acceleration, passing three points A, B and C.
It takes 6 seconds to travel from A to B and a further 4 seconds to travel 10.4 metres from B to C.
It is travelling at 5 ms^{-1} as it passes point B. Find

a) the acceleration of the car.

b) the speed of the car at A.

c) how far the car travels beyond point C before it comes to rest.

Question 15. A vehicle travels with constant acceleration along a straight flat road.
It passes three checkpoints A, B and C and in each case is travelling at 15 ms^{-1}, 30 ms^{-1} and 50 ms^{-1}.
Find

a) the ratio of the times taken to drive the distances AB and BC.
b) the distance between AB given the distance AC is 150 metres.

Question 16. A truck is travelling along a straight horizontal road with constant acceleration.
A set of traffic lights is 25 metres away from the truck. A junction is a further 110 metres on from the traffic lights. The truck takes 2 seconds to travel to the traffic lights and 5 seconds to travel between the traffic lights and the junction.
Find

a) the acceleration of the truck.
b) the speed of the truck in its initial position.

Question 17. Two cars, A and B, are travelling along a flat, straight road.
Car A is driving at u ms^{-1} throughout its motion.
Initially, car B is 15 metres behind car A and is driving 0.2 ms^{-1} slower than car A but is accelerating at 0.8 ms^{-2}.
Find the time it takes for Car B to catch up Car A.

Question 18. A car is travelling on a straight flat road.
Initially it has velocity of 6 ms^{-1} and drives with constant acceleration, a ms^{-2}, past two checkpoints.
It reaches the first checkpoint which is at 48 metres from the initial point, in k seconds.

a) Find a giving your answer in terms of k
b) Find the initial speed for the second phase of the motion in terms of k

The next checkpoint is a further 204 metres away and it reaches it in 10 seconds.

c) Find the value of k
d) Find the value of a

Question 19. A stone is dropped on a planet from a height of 3 metres. It takes 2.2 seconds to hit the surface. Find the acceleration due to gravity on this planet.

Question 20. The following questions use $g = 9.81$ $\mathbf{ms}^{-2}$ A ball is dropped, and it takes 0.8 seconds to reach the ground. Find

a) the height above the ground from which the ball is dropped.
b) the speed in terms of t
c) the speed when the ball hits the ground.

Question 21. The following questions use $g = 9.8$ **ms**$^{-2}$ An object, which is initially at rest is dropped from a height of 8m above the ground.

a) Calculate the speed of the object when it hits the ground.
b) Calculate the time it takes to hit the ground.
c) State a refinement to the model, apart from air resistance, which would make the model more realistic.

Question 22. A ball is projected upwards from the ground with speed 14.7 ms^{-1}. Find

a) the time it takes to reach ground again.
b) the total distance travelled in the first 2 seconds of motion.

Question 23. A pebble is thrown upwards in the air from the edge of a bridge with a velocity of 24.5 ms^{-1}.

a) Find the time it takes for the ball to reach its maximum height.

After 6.5 seconds from being thrown, the pebble lands in the river below. Find

b) the height of the bridge at the point where the pebble was thrown.

c) the total distance travelled by the pebble.

Question 24. A stone is thrown vertically upwards in the air starting 1.65 m above the ground being projected with initial speed 6.25 ms^{-1}. The stone moves freely under gravity.

a) Find the time at which the stone hits the ground.

b) Find the times at which the stone is precisely 2 m above the ground.

c) Hence find the total time the stone spends more than 2 metres above the ground.

Question 25. The following questions use $g = 10$ **ms**$^{-2}$ A ball is thrown vertically upwards with initial speed u ms^{-1} from a height of 15m above the ground. It hits the ground with speed 22 ms^{-1}

a) Find the initial speed.

b) What is the greatest height it reaches?

c) Find the time it takes to hit the ground.

d) Given the ball should be subject to air resistance, explain the how this would affect your answers to the previous parts of the question.

Question 26. A parachutist steps off a plane at a height of 2800 m. They free-fall for 4 seconds before opening their parachute.

a) Find the speed of the parachutist the moment the parachute opens.

It is given that from the point the parachute opens, they decelerate at 12 ms^{-2} for 3 seconds before reaching a constant speed.

b) Find the total time it takes to reach the ground from initially stepping off the plane.

Question 27. A ball is thrown vertically upwards from the ground at 20 ms^{-1} whilst simultaneously a second identical ball is thrown vertically downwards at 5 ms^{-1} from a height of 10 metres above the ground. Find

a) the time at which both balls are at the same height.
b) the distance from the ground at which both balls are at the same height.

Question 28. A ball A is thrown vertically upwards from the ground at 20 ms^{-1} whilst simultaneously a second identical ball B is thrown vertically upwards at 12 ms^{-1} from a height of 5 metres above the ground. Find

a) the amount of time that both balls are above the height of the initial start point of ball B.
b) the time at which both balls are at the same height.
c) the distance from the ground at which both balls are at the same height.

Question 29. The following questions use g in exact form An object is dropped from a height of h metres. Show that $v = \sqrt{2gh}$

Question 30. A stone is dropped from a height of h metres above horizontal ground. It falls freely downwards. Its speed when reaching the ground is 8 ms^{-1}. Show that $h = \frac{32}{g}$

Question 31. A ball is projected directly upwards at u ms^{-1}. Find an expression for

a) the maximum height in terms of u

b) the time it takes to return to its start point in terms of u

Question 32. Two particles A and B are released from rest. A is released from k metres above a horizontal surface. B is released 0.5 seconds later from $\frac{k}{10}$ metres above the horizontal surface. They land simultaneously on the surface at t seconds after the release of A.

Find the value of t and show that $k \approx 0.267g$